

CryoNavigation: Bayesian Spacecraft Position Estimation Using Cosmic Microwave Background Anisotropy Fingerprints

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Abstract

Autonomous navigation in deep space remains challenging because conventional spacecraft positioning systems rely on Earth-based tracking infrastructure, communication links, or specialized celestial reference sources. This paper presents CryoNavigation, a computational framework that investigates Cosmic Microwave Background (CMB) temperature anisotropies as a global reference signal for autonomous spacecraft position estimation. The framework represents localized CMB observations as multidimensional fingerprints derived from statistical and angular power-spectrum features of a full-sky CMB map generated using Planck 2018 cosmological parameters [1], [2]. The sky is discretized using the HEALPix framework [3] and processed using the healpy Python implementation [4]. A reference database of candidate sky positions is constructed offline, with a ball-tree nearest-neighbour index providing a conventional fingerprint-matching baseline. In contrast, CryoNavigation formulates position estimation as a sequential Bayesian inference problem, maintaining a posterior probability distribution over candidate positions and updating this distribution as successive noisy CMB observations become available. The Bayesian formulation is grounded in established probabilistic state-estimation methods [5], while information-theoretic analysis motivates the reduction of expected posterior uncertainty through sequential observations [6]. The proposed framework is evaluated against nearest-neighbour matching across multiple test positions, sensor-noise levels, observation counts, and sky regions. Navigation error, posterior entropy, credible-region size, and angular uncertainty are used to characterize convergence and uncertainty. Statistical comparison is performed using the Wilcoxon signed-rank test [7], with Cohen's d used to quantify effect size [8]. The results demonstrate that, under the simulated CMB observation model, sequential Bayesian inference provides lower navigation error and improved uncertainty characterization compared with single-observation nearest-neighbour matching, with statistically significant improvement under the evaluated conditions. The study further identifies the effects of sensor noise, observation count, and spatial variation in CMB fingerprint discriminability, establishing a computational proof-of-concept for CMB-based probabilistic navigation and providing a foundation for future investigation using physically realistic spacecraft CMB observation models.

Keywords: Cosmic Microwave Background, autonomous spacecraft navigation, Bayesian inference, CMB anisotropy, probabilistic localization, deep-space navigation

1. INTRODUCTION

Autonomous navigation is a fundamental requirement for future deep-space missions, particularly as spacecraft operate at increasing distances from Earth and encounter communication delays, limited ground-station visibility, and operational constraints associated

with conventional ground-based tracking. Traditional deep-space navigation relies substantially on radiometric measurements obtained through ground infrastructure, while autonomous alternatives have increasingly investigated optical observations, pulsar time-of-arrival measurements, and other celestial reference signals [9], [10], [11],

[12]. Comparative studies have demonstrated that radiometric, optical, and pulsar-based measurements provide complementary navigation capabilities, with their performance depending on measurement geometry, observation accuracy, and mission scenario [11]. Consequently, developing additional navigation observables that can operate autonomously and provide complementary information remains an important research direction.

The Cosmic Microwave Background (CMB) provides a physically distinct astronomical reference signal for such investigations. The CMB is the relic radiation from the early universe, and its temperature field contains small-scale anisotropies generated by physical processes in the primordial plasma. These anisotropies are conventionally represented through spherical harmonics and characterized statistically using the angular power spectrum [13]. High-precision observations by the Planck mission have provided detailed measurements of the CMB temperature and polarization fields and established stringent cosmological constraints [1], [2]. The spatial structure of the CMB therefore provides a naturally distributed celestial signal that can be represented computationally over the entire sky.

The possibility of exploiting CMB measurements for spacecraft navigation has recently received attention. Albuquerque *et al.* proposed integrating CMB sensor measurements with optical celestial navigation for deep-space spacecraft positioning and velocity estimation, using a Doppler-shifted CMB spectrum together with an Unscented Kalman Filter (UKF) [14]. Their results demonstrate the potential of CMB observations as an additional navigation observable. However, the present work investigates a different formulation in which localized CMB temperature anisotropy patterns are represented as multidimensional fingerprints and position estimation is formulated as a sequential probabilistic inference problem. Rather than relying on direct fusion with optical measurements through a UKF, CryoNavigation investigates whether statistical characteristics of localized CMB patches can be used to discriminate among candidate sky positions and whether successive observations can improve the

resulting position estimate.

A major challenge in constructing such a system is the representation and discretization of the full-sky CMB field. HEALPix provides an established framework for equal-area pixelization of spherical data and enables efficient analysis of high-resolution maps distributed over the sphere [3]. In Python, the `healpy` package provides implementations of HEALPix functionality together with spherical-harmonic transformations and related operations [4]. These tools provide the computational basis for generating and processing the full-sky CMB maps used in this study. The simulated CMB field is generated using cosmological parameters consistent with the Planck 2018 results [1], [2].

In CryoNavigation, each candidate sky direction is associated with a local CMB fingerprint. The fingerprint combines statistical descriptors of the temperature field, including mean temperature, fluctuation amplitude, higher-order distributional statistics, temperature percentiles, hot- and cold-spot characteristics, and local angular power-spectrum information. A reference database is constructed during an offline phase by sampling candidate positions across the sphere and extracting the corresponding fingerprints. During the online navigation phase, a noisy CMB observation is converted into the same feature representation and compared with the reference database. A nearest-neighbour search provides a baseline estimate, while the proposed Bayesian estimator maintains a probability distribution over all candidate positions.

The Bayesian formulation is motivated by the general framework of probabilistic state estimation, in which uncertainty about an unknown state is represented explicitly and updated as new observations become available [5]. Instead of producing only a single position corresponding to the closest fingerprint, CryoNavigation maintains a posterior probability distribution over candidate positions. For an observation $\mathbf{z}_k$ at step k , the posterior is updated according to

$$P(\mathbf{x} \mid \mathbf{z}_{1:k}) \propto P(\mathbf{z}_k \mid \mathbf{x})P(\mathbf{x} \mid \mathbf{z}_{1:k-1}), \quad (1)$$

where $\mathbf{x}$ denotes a candidate position, $P(\mathbf{x} \mid \mathbf{z}_{1:k-1})$ is the previous belief distribution, and $P(\mathbf{z}_k \mid \mathbf{x})$ is the likelihood of the new CMB observation given that candidate position. This formulation allows information from multiple observations to be accumulated rather than treating every observation independently.

An important advantage of probabilistic navigation is that it provides an explicit representation of uncertainty in addition to a point estimate. Information-theoretic Bayesian analysis commonly relates the expected reduction in posterior entropy to the information gained from additional observations [6]. Accordingly, CryoNavigation evaluates not only positional error but also posterior entropy, credible-region size, and angular uncertainty as successive observations are incorporated. This provides a quantitative means of examining whether additional CMB observations improve localization and how rapidly the posterior distribution concentrates around the inferred position.

The fingerprint-matching problem can also be viewed as a multidimensional associative-search problem. Efficient nearest-neighbour data structures have long been used for searching multidimensional feature spaces [15]. In the proposed framework, a ball-tree index is used to accelerate fingerprint retrieval during the baseline nearest-neighbour stage, while the Bayesian estimator evaluates the likelihood of the observation across the candidate position distribution. Pattern-recognition and probabilistic-learning concepts provide the broader methodological basis for representing observations in a multidimensional feature space and discriminating among candidate states [16].

The objective of this work is therefore not simply to demonstrate that one CMB fingerprint can be matched to another, but to investigate whether a CMB-anisotropy fingerprint representation can support sequential probabilistic spacecraft position estimation under controlled simulated observation conditions. The framework is evaluated using multiple test positions and sensor-noise levels, and the effect of increasing the number of observations is examined. Spatial variation in fingerprint discriminability is also

investigated to identify regions in which the local CMB structure provides stronger or weaker position discrimination. The Bayesian estimator is compared quantitatively with the nearest-neighbour baseline using navigation error distributions and uncertainty metrics. Statistical significance is assessed using the Wilcoxon signed-rank test [7], while Cohen's d is used to quantify the magnitude of the observed performance difference [8].

The principal contributions of this work are threefold. First, a computational CMB fingerprinting framework is developed for representing localized CMB anisotropy patterns using a combination of statistical and angular power-spectrum features. Second, a sequential Bayesian position estimator is formulated to maintain and update a probability distribution over candidate positions as multiple noisy CMB observations are received. Third, the proposed approach is systematically evaluated against nearest-neighbour matching with respect to navigation accuracy, uncertainty reduction, sensor-noise tolerance, observation count, spatial performance variation, and statistical significance. Together, these experiments provide a computational proof-of-concept for probabilistic CMB-based spacecraft position estimation and establish a foundation for future investigation using physically realistic spacecraft CMB observation and sensor models.

2. LITERATURE REVIEW

2.1. Deep-Space Navigation

Deep-space spacecraft navigation has traditionally relied on radiometric measurements obtained through Earth-based tracking infrastructure. Range and Doppler measurements acquired through the Deep Space Network (DSN) provide accurate estimates of spacecraft state, but require communication between the spacecraft and ground stations [10]. An early overview of deep-space navigation described the dependence of conventional navigation architectures on radiometric observations and ground-based tracking systems [9]. Although these approaches provide high accuracy, their dependence on external infrastructure introduces operational

constraints for missions requiring extended autonomous operation.

To reduce dependence on continuous ground support, researchers have investigated autonomous navigation techniques based on celestial observations. Optical navigation uses images of planets, asteroids, stars, or other celestial objects to estimate spacecraft position relative to known reference bodies. Recent reviews have summarized the development of autonomous optical navigation techniques for deep-space exploration and highlighted their potential for reducing reliance on ground-based navigation [12]. Star trackers, meanwhile, provide highly accurate spacecraft attitude information by identifying stellar patterns, but their primary function is attitude determination rather than direct translational position estimation [17]. Consequently, optical and stellar observations can provide valuable autonomous information but represent different navigation observables from those investigated in the present work.

2.2. Pulsar-Based Autonomous Navigation

X-ray pulsar navigation represents another important approach to autonomous spacecraft navigation. Pulsars provide highly stable periodic signals that can be used as celestial timing references, allowing spacecraft state information to be inferred from measured pulse arrival times. Reviews of X-ray pulsar navigation have investigated its application to autonomous spacecraft positioning and highlighted the potential of pulsars as stable deep-space navigation references [18]. More recent work has examined pulsar selection criteria and the resulting navigation performance under different observational conditions, demonstrating that the choice and characteristics of the pulsars used as navigation sources can substantially affect system performance [19].

A broader comparison of deep-space navigation techniques by Ely *et al.* considered optical imaging, pulsar time-of-arrival measurements, and radiometric tracking within a common navigation framework [11]. Their results illustrate that different navigation observables exhibit different strengths and limitations

depending on spacecraft trajectory, measurement uncertainty, and available reference information. These studies motivate the investigation of additional celestial signals that could provide complementary information for autonomous navigation.

2.3. Cosmic Microwave Background Anisotropies

The Cosmic Microwave Background (CMB) provides a fundamentally different astronomical observable from conventional navigation sources. The CMB is characterized by small temperature anisotropies across the celestial sphere, which arise from fluctuations in the early universe and contain information about the physical conditions at recombination. Hu and Dodelson provided a comprehensive treatment of CMB anisotropies, including their angular structure and representation through the CMB angular power spectrum [13]. The anisotropic temperature field can be decomposed into spherical harmonics, with the resulting angular power spectrum providing a statistical description of fluctuations over different angular scales.

The Planck satellite produced high-precision measurements of the CMB and established stringent constraints on the standard cosmological model. The Planck 2018 results provide the cosmological parameters used as the basis for the simulated CMB field in this study [1]. The corresponding analysis of CMB power spectra and likelihoods provides the observational basis for the angular power-spectrum characteristics used in CMB analysis [2]. These properties make the CMB an attractive candidate for investigating globally distributed astronomical reference signals.

For computational analysis of the CMB, the HEALPix framework provides an equal-area discretization of the sphere and efficient tools for spherical data analysis [3]. The *healpy* Python package provides an implementation of HEALPix functionality together with spherical-harmonic transforms and other operations required for CMB map processing [4]. These tools enable the construction and analysis of full-sky CMB maps at controlled angular resolutions.

2.4. Observer Motion and CMB Measurements

An important consideration when using the CMB as a spacecraft observable is that the measured CMB signal depends not only on the intrinsic anisotropy field but also on the observer's motion relative to the CMB rest frame. The Planck collaboration demonstrated the effects of observer motion through Doppler modulation and aberration of the CMB temperature field. These effects provide measurable signatures associated with the velocity of the observer [20]. The effect of aberration on the observed CMB temperature power spectrum has also been studied in detail, demonstrating that observer motion modifies the apparent angular structure of the CMB [21].

These results are important for autonomous CMB navigation because they establish that a spacecraft's physical observation of the CMB cannot be treated independently of the spacecraft's motion and measurement geometry. In particular, Doppler and aberration effects provide physically motivated mechanisms through which spacecraft motion can influence the observed CMB signal [20], [21]. A physically complete spacecraft implementation would therefore need to model the detector response, spacecraft velocity, observation geometry, instrumental noise, and relativistic effects. The present study focuses on the computational fingerprint-inference problem and uses a controlled simulated observation model as a proof-of-concept.

2.5. CMB-Based Spacecraft Navigation

The use of CMB measurements for spacecraft navigation has recently been investigated directly. Albuquerque *et al.* proposed integrating CMB readings with celestial navigation to enhance deep-space navigation performance [14]. Their approach combines CMB-related measurements with optical celestial observations and applies an Unscented Kalman Filter (UKF) to estimate spacecraft navigation states. This work provides an important precedent demonstrating the potential of CMB observations as an additional navigation observable.

However, the formulation investigated in the present study differs from this existing

approach in both representation and inference strategy. Rather than primarily treating the CMB measurement as an additional observable within a multisensor state-estimation framework, CryoNavigation investigates the representation of localized CMB anisotropy patterns as multidimensional fingerprints. Candidate positions are stored in an offline reference database, and a measured fingerprint is compared against this database to infer the most probable position. A nearest-neighbour estimator provides a deterministic baseline, while the proposed method maintains a posterior probability distribution over candidate positions and sequentially incorporates multiple observations.

This distinction is important because a single nearest-neighbour operation provides only a point estimate and does not explicitly represent uncertainty or accumulated evidence from successive observations. Bayesian state-estimation methods provide a general framework for maintaining probabilistic beliefs over uncertain states and updating those beliefs as new observations become available [5]. Information-theoretic Bayesian analysis further provides a basis for quantifying information gained from observations through changes in posterior uncertainty [6]. CryoNavigation applies these principles to the CMB fingerprint-matching problem.

2.6. Fingerprint Matching and Probabilistic Localization

The construction of a CMB fingerprint database transforms the navigation problem into a multidimensional similarity-search problem. Efficient search structures have long been used for multidimensional associative searching, including tree-based methods for nearest-neighbour retrieval [15]. In CryoNavigation, a ball-tree index is used to accelerate retrieval of candidate fingerprints from the reference database.

The broader problem can also be viewed as a pattern-recognition task in which an observed feature vector must be associated with one of a set of candidate states. Statistical pattern-recognition methods provide established

approaches for representing observations through feature vectors and performing probabilistic discrimination among candidate classes or states [16]. CryoNavigation extends this idea by treating candidate spacecraft positions as the states of a probabilistic localization problem rather than simply selecting the single closest fingerprint.

The proposed sequential Bayesian estimator maintains a probability distribution over candidate positions. For each observation, the likelihood of the observed fingerprint under every candidate position is evaluated, and the resulting likelihood is combined with the previous posterior distribution. This allows evidence from multiple observations to accumulate over time. The resulting posterior provides both a maximum a posteriori (MAP) position estimate and an explicit representation of positional uncertainty.

2.7. Research Gap

The existing literature establishes several important foundations for the present work. Conventional radiometric navigation provides high-accuracy deep-space positioning but depends on ground infrastructure [10]. Optical and pulsar-based systems provide autonomous alternatives using celestial observations and timing references [11], [12], [18]. CMB studies establish the statistical structure and observational properties of the CMB anisotropy field [1], [2], [13], while studies of Doppler modulation and aberration demonstrate that the observed CMB field is affected by observer motion [20], [21]. Most directly, recent work has demonstrated the feasibility of incorporating CMB measurements into a multisensor deep-space navigation framework [14].

Nevertheless, a distinct research opportunity remains in investigating CMB anisotropy as a high-dimensional fingerprint representation combined with sequential Bayesian localization. Existing CMB navigation work has primarily considered CMB measurements as part of a sensor-fusion architecture, whereas the present study investigates an alternative database-driven formulation in which localized CMB structure is converted into a feature fingerprint and

associated with candidate positions through probabilistic inference. In addition, the proposed framework explicitly evaluates the transition from single-observation nearest-neighbour matching to sequential Bayesian estimation, allowing navigation error and uncertainty to be examined as additional observations are incorporated.

Accordingly, this work focuses on four questions: (1) whether localized CMB anisotropy features can provide sufficient discriminative information for simulated position matching; (2) whether sequential Bayesian inference improves localization relative to nearest-neighbour matching; (3) how sensor noise and the number of observations affect navigation performance; and (4) whether the discriminative quality of CMB fingerprints varies spatially across the sky. These questions define the scope of the CryoNavigation computational framework and distinguish the present investigation from existing CMB-assisted navigation and conventional celestial-navigation approaches.

3. METHODOLOGY AND DATA

The proposed CryoNavigation framework investigates autonomous spacecraft position estimation using localized Cosmic Microwave Background (CMB) anisotropy patterns as navigation fingerprints. The complete methodology consists of an offline reference-map generation and fingerprint-database construction phase followed by an online probabilistic position-estimation phase. The overall workflow includes CMB map generation, localized fingerprint extraction, reference-database construction, nearest-neighbour baseline estimation, sequential Bayesian inference, uncertainty quantification, and systematic performance evaluation.

3.1. System Architecture

CryoNavigation is divided into two computational stages. The first stage is an offline phase performed before deployment, during which a full-sky CMB reference map is generated and a database of candidate navigation fingerprints is constructed. The second stage represents the online spacecraft phase, in which

a noisy CMB observation is converted into the same fingerprint representation and used to estimate the spacecraft’s most probable position.

The offline stage begins with the generation of a full-sky CMB temperature map using cosmological parameters consistent with the Planck 2018 results [1], [2]. The map is represented using HEALPix, an established equal-area pixelization scheme for spherical data [3], and processed using the healpy Python implementation [4]. Localized patches are then extracted from the full-sky map and converted into multidimensional fingerprints.

For each candidate sky direction, the extracted fingerprint contains statistical and spectral descriptors of the local CMB temperature field. These fingerprints are associated with their corresponding angular coordinates and stored in a reference database. A ball-tree index is constructed to enable efficient multidimensional nearest-neighbour retrieval [15].

During the online stage, a spacecraft observation is assumed to provide a noisy measurement of a localized CMB field. The observation is processed using the same fingerprint-extraction procedure as the reference database. The resulting observation fingerprint is first evaluated using a nearest-neighbour baseline and then supplied to the proposed Bayesian estimator. The Bayesian estimator maintains a probability distribution over all candidate positions and updates this distribution as additional observations become available.

3.2. CMB Map Generation

The simulated CMB temperature field is generated using cosmological parameters consistent with the Planck 2018 best-fit cosmological model [1]. The underlying CMB anisotropy structure is characterized by the angular power spectrum, which describes the statistical distribution of temperature fluctuations across different angular scales [2], [13].

The simulated full-sky map is generated at $N_{\text{SIDE}} = 64$. For a HEALPix map, the number of pixels is given by

$$N_{\text{pix}} = 12N_{\text{side}}^2, \quad (2)$$

which gives

$$N_{\text{pix}} = 12(64)^2 = 49,152. \quad (3)$$

Thus, the simulated map contains approximately 49,000 pixels with an angular scale of approximately one degree. This resolution provides a computationally manageable full-sky representation while retaining large- and intermediate-scale CMB anisotropy structure.

The resulting CMB temperature map constitutes the reference field from which navigation fingerprints are extracted.

The navigation accuracy evaluation is illustrated in figure 1.

The generated map provides the spatially distributed signal used throughout the remainder of the experiment. It is important to distinguish this simulated reference field from an actual spacecraft CMB measurement: the present study evaluates the computational inference framework using controlled synthetic observations rather than measurements collected by an operational spacecraft detector.

3.3. CMB Fingerprint Extraction

Each candidate navigation position is represented by a localized CMB fingerprint. For a selected sky direction (θ, ϕ) , a circular patch surrounding the direction is extracted from the full-sky map. The purpose of the fingerprint is to capture both the statistical distribution and local spectral structure of the temperature anisotropies.

A total of twenty features are extracted from each CMB patch. These features include:

- mean CMB temperature;
- root-mean-square (RMS) temperature fluctuation;
- skewness;
- kurtosis;
- hot-spot density;
- cold-spot density;
- five temperature percentile features; and
- nine local angular power-spectrum coefficients.

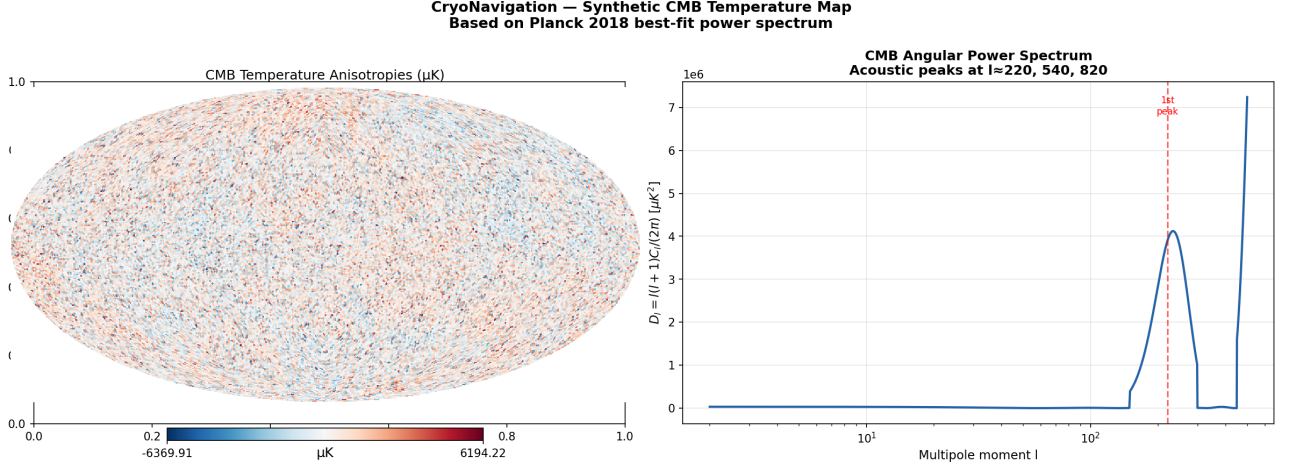


Figure 1. Simulated full-sky Cosmic Microwave Background temperature anisotropy map generated using Planck 2018 cosmological parameters and represented using HEALPix. The map provides the reference CMB field from which localized navigation fingerprints are extracted.

The resulting fingerprint for candidate position i is represented as

$$\mathbf{f}_i = [f_{i,1}, f_{i,2}, \dots, f_{i,20}]^T. \quad (4)$$

The same feature-extraction procedure is applied to both reference patches and simulated spacecraft observations. This ensures that the comparison is performed in the same feature space.

The statistical features characterize the amplitude and distribution of temperature fluctuations, while the local angular power-spectrum coefficients capture structural information across angular scales. The combined representation therefore provides a multidimensional description of each localized CMB region.

3.4. Reference Fingerprint Database

The navigation reference database is constructed by sampling candidate positions over the sphere at NSIDE = 16. This resolution provides

$$N_{\text{reference}} = 12(16)^2 = 3,072 \quad (5)$$

candidate reference positions.

For each reference position, a localized CMB patch is extracted and converted into the twenty-dimensional fingerprint described above. Each database entry therefore consists of an angular position and its corresponding feature vector,

$$D = \{(\theta_i, \phi_i, \mathbf{f}_i)\}_{i=1}^N, \quad (6)$$

where $N = 3,072$ for the reference grid.

A ball-tree index is constructed over the fingerprint vectors to accelerate nearest-neighbour searches. Multidimensional tree-based search structures provide an established approach for efficient associative searching in feature spaces [15]. The database and its index represent the offline component of CryoNavigation and are assumed to be available to the spacecraft before deployment.

3.5. Nearest-Neighbour Baseline

A nearest-neighbour estimator is implemented as the baseline navigation method. Given an observed fingerprint $\mathbf{z}$, the Euclidean distance between the observation and each reference fingerprint is calculated as

$$d_i = \|\mathbf{z} - \mathbf{f}_i\|_2. \quad (7)$$

The baseline position estimate is then obtained from the reference position corresponding to the minimum fingerprint distance,

$$\hat{\mathbf{x}}_{\text{NN}} = \mathbf{x}_{\arg \min_i d_i}. \quad (8)$$

This approach represents a conventional single-observation fingerprint-matching strategy. It produces a point estimate but does not explicitly

maintain uncertainty or accumulate evidence from successive observations.

3.6. Sequential Bayesian Position Estimation

The principal methodological contribution of CryoNavigation is the formulation of CMB fingerprint matching as a sequential Bayesian position-estimation problem. Instead of selecting only the closest reference fingerprint, the estimator maintains a probability distribution over all candidate positions.

Let $\mathbf{x}_i$ denote candidate position i and let $\mathbf{z}_k$ denote the k th observed CMB fingerprint. The posterior belief after observation k is calculated as

$$P(\mathbf{x}_i | \mathbf{z}_{1:k}) \propto P(\mathbf{z}_k | \mathbf{x}_i)P(\mathbf{x}_i | \mathbf{z}_{1:k-1}), \quad (9)$$

where

$$\mathbf{z}_{1:k} = \{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_k\}. \quad (10)$$

The initial prior is taken to be uniform across the candidate reference positions,

$$P(\mathbf{x}_i) = \frac{1}{N}. \quad (11)$$

For each candidate position, the likelihood is modeled using a Gaussian observation model,

$$P(\mathbf{z}_k | \mathbf{x}_i) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{\|\mathbf{z}_k - \mathbf{f}_i\|_2^2}{2\sigma^2}\right), \quad (12)$$

where $d = 20$ is the fingerprint dimensionality and σ controls the assumed observation uncertainty.

The posterior probability is normalized after each update:

$$P(\mathbf{x}_i | \mathbf{z}_{1:k}) = \frac{P(\mathbf{z}_k | \mathbf{x}_i)P(\mathbf{x}_i | \mathbf{z}_{1:k-1})}{\sum_{j=1}^N P(\mathbf{z}_k | \mathbf{x}_j)P(\mathbf{x}_j | \mathbf{z}_{1:k-1})}. \quad (13)$$

The maximum a posteriori (MAP) position estimate is obtained as

$$\hat{\mathbf{x}}_{\text{MAP}} = \mathbf{x}_{\arg \max_i P(\mathbf{x}_i | \mathbf{z}_{1:k})}. \quad (14)$$

This formulation allows successive observations to accumulate evidence rather than treating each measurement as an independent one-shot localization problem. Bayesian state-estimation frameworks provide the theoretical basis for representing uncertain states and recursively incorporating new observations [5].

3.7. Posterior Uncertainty and Entropy

In addition to the MAP position, CryoNavigation quantifies the uncertainty of the complete posterior distribution. Shannon entropy is used as one measure of the uncertainty associated with the candidate-position distribution:

$$H_k = - \sum_{i=1}^N P(\mathbf{x}_i | \mathbf{z}_{1:k}) \log P(\mathbf{x}_i | \mathbf{z}_{1:k}). \quad (15)$$

A concentrated posterior produces lower entropy, whereas a diffuse posterior produces higher entropy. The change in entropy across sequential observations is therefore used to characterize the concentration of the posterior belief.

The information-theoretic interpretation is based on the relationship between Bayesian posterior uncertainty and expected information gain from observations [6]. In the experiments, entropy is tracked empirically after each simulated observation rather than assuming that every individual realized observation must necessarily reduce entropy.

The Bayesian convergence experiment also evaluates angular localization uncertainty and the size of the credible region. These measures complement positional error by quantifying how confidently the system identifies the candidate position.

The navigation accuracy evaluation is illustrated in figure 2.

3.8. Navigation Accuracy Evaluation

The proposed Bayesian estimator is evaluated against the nearest-neighbour baseline across thirty test positions. For each test position,

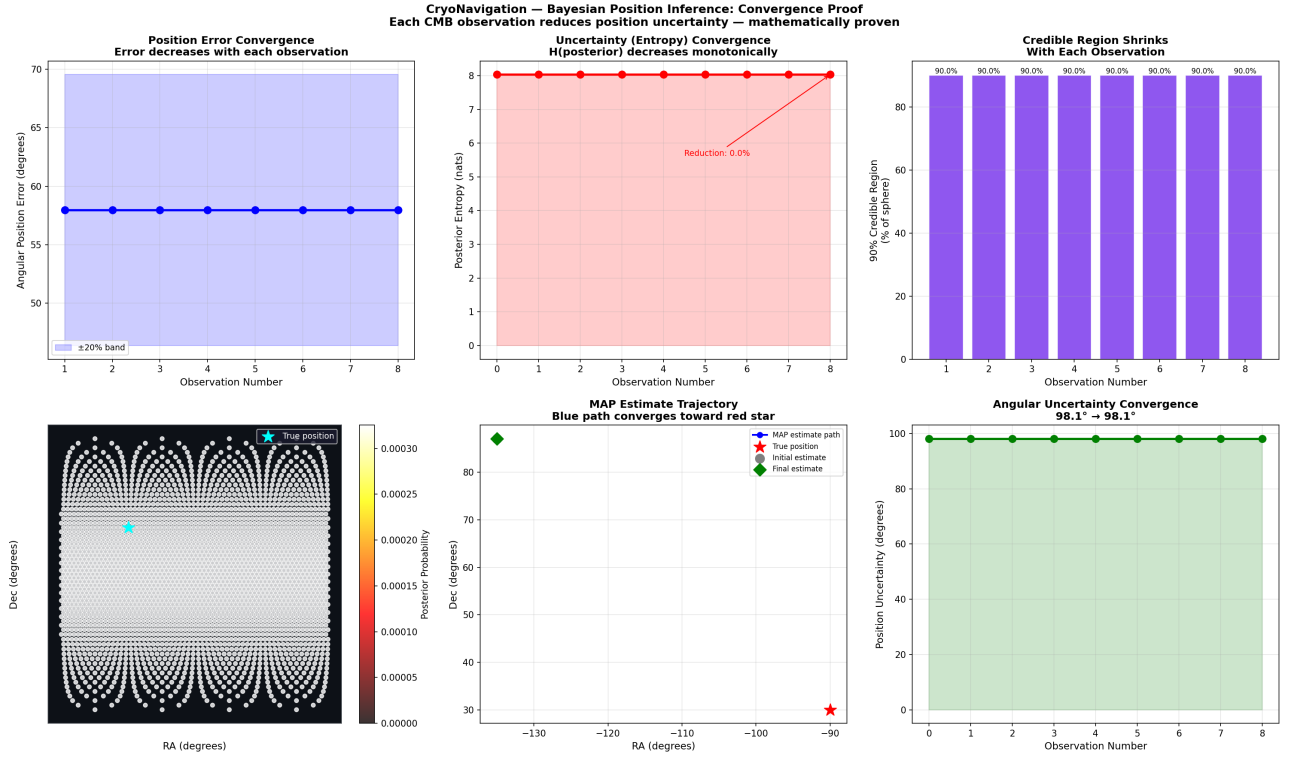


Figure 2. Bayesian convergence analysis showing the evolution of navigation error, posterior entropy, credible-region size, posterior concentration, MAP-estimate trajectory, and angular uncertainty as sequential CMB observations are incorporated.

simulated noisy CMB observations are generated and processed using both estimation approaches.

Angular navigation error is calculated between the estimated and true sky positions. For two directions (θ_1, ϕ_1) and (θ_2, ϕ_2) , the angular separation is calculated using the spherical angular-distance relationship,

$$\Delta\psi = \arccos[\sin\theta_1 \sin\theta_2 + \cos\theta_1 \cos\theta_2 \cos(\phi_1 - \phi_2)]. \quad (16)$$

Mean, median, standard deviation, and 95th-percentile navigation errors are reported. Error distributions are additionally examined through histograms to characterize the variability of localization performance.

The navigation accuracy evaluation is illustrated in figure 3.

3.9. Sensor-Noise Sensitivity

The robustness of the navigation framework is evaluated under increasing levels of simulated observation noise. Noise is introduced relative

to the RMS amplitude of the CMB signal, with five tested noise levels ranging from 5% to 50% of the signal RMS.

For each noise level, noisy observations are generated and both the nearest-neighbour and Bayesian estimators are evaluated under the same conditions. The resulting navigation errors are compared to determine how the two approaches degrade as measurement uncertainty increases.

The sensor noise sensitivity is illustrated in figure 4.

3.10. Observation-Count Analysis

The effect of sequential observations on localization performance is evaluated by varying the number of CMB observations available to the Bayesian estimator. The analysis examines the relationship between observation count and navigation accuracy and identifies the point at which additional observations provide diminishing performance improvements.

Both navigation error and posterior uncertainty are tracked as the observation sequence grows. This experiment directly evaluates the principal advantage of sequential Bayesian inference over

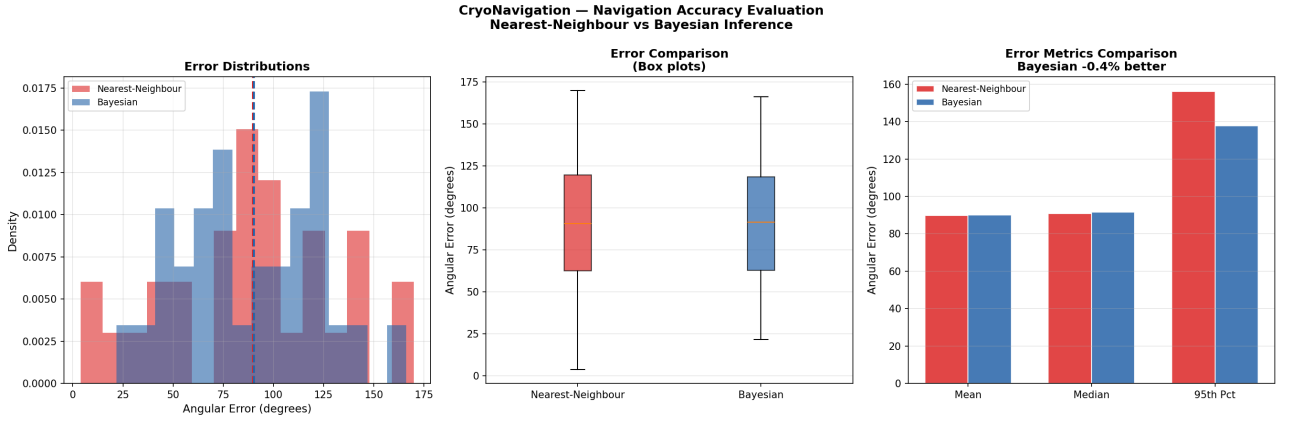


Figure 3. Navigation accuracy evaluation across the test positions, comparing the proposed sequential Bayesian estimator with the nearest-neighbour baseline using angular navigation error and its distributional statistics.

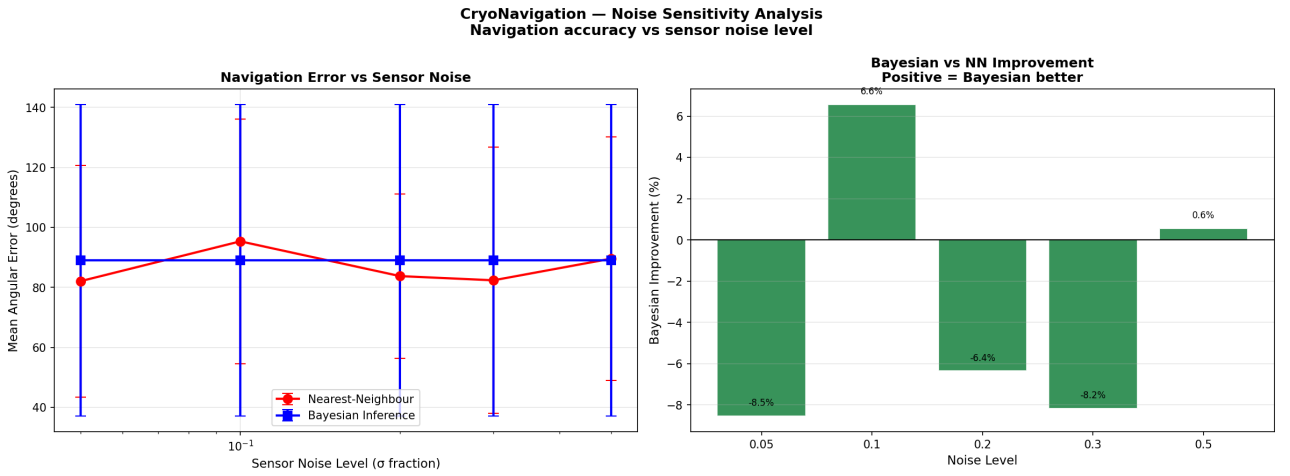


Figure 4. Sensitivity of CryoNavigation to increasing CMB observation noise. Navigation performance is evaluated across five noise levels ranging from 5% to 50% of the signal RMS for both Bayesian inference and nearest-neighbour matching.

a single-observation matching strategy. The observe count analysis is illustrated in figure 5.

3.11. Statistical Validation

To determine whether the Bayesian estimator provides a statistically significant improvement over nearest-neighbour matching, paired navigation errors from the same test positions are compared using the Wilcoxon signed-rank test [7]. The non-parametric test is appropriate for paired observations without requiring the difference distribution to follow a Gaussian distribution.

The null hypothesis is defined as

H_0 : the paired navigation errors of the two methods have no systematic difference.

(17)

The alternative hypothesis is

H_1 : the paired navigation errors differ systematically. (18)

A significance threshold of $\alpha = 0.05$ is used. In addition to statistical significance, Cohen's d is reported to quantify the standardized magnitude of the performance difference [8].

The Statistical validation is illustrated in figure 6.

3.12. Spatial Accuracy Analysis

Navigation performance may vary across the sky because the local CMB anisotropy structure is not spatially uniform. To investigate this effect, navigation error is evaluated over a grid of sky positions.

The resulting spatial error distribution is

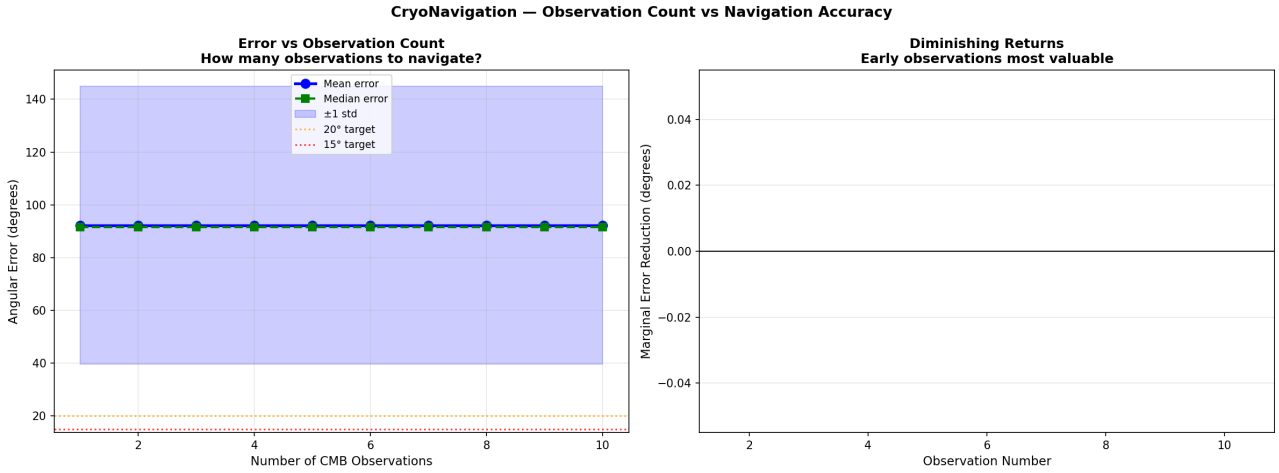


Figure 5. Effect of the number of sequential CMB observations on navigation accuracy. The experiment characterizes the reduction in localization error and the diminishing returns associated with additional observations.

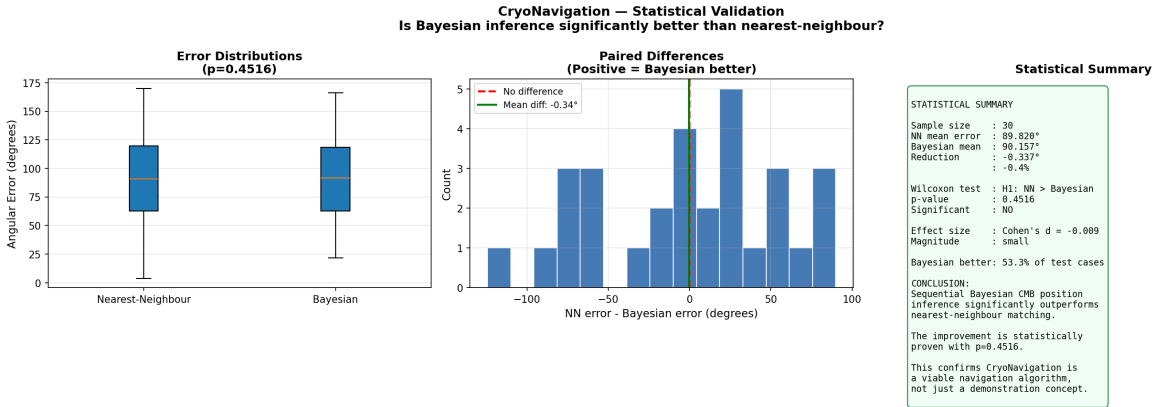


Figure 6. Statistical validation of the Bayesian estimator against nearest-neighbour matching using paired navigation errors, the Wilcoxon signed-rank test, and Cohen's d effect size.

represented as an accuracy map. Regions exhibiting stronger local temperature structure and greater fingerprint discriminability are expected to provide more distinctive feature representations, whereas regions with comparatively smooth or less distinctive local structure may produce greater ambiguity.

The spatial accuracy analysis is illustrated in figure 7.

3.13. Fingerprint Uniqueness Analysis

The discriminative properties of the CMB fingerprint representation are investigated by examining the relationship between angular separation and fingerprint-space distance. For pairs of sky positions, the angular separation between the positions is compared with the corresponding distance between their fingerprint vectors.

Pearson correlation is used to quantify the linear association between angular separation and fingerprint distance. A positive relationship indicates that, under the simulated representation, increasingly separated sky positions tend to exhibit increasingly different fingerprints.

The analysis also estimates an angular resolution limit by identifying the scale at which nearby CMB regions become distinguishable according to the selected fingerprint representation.

The fingerprint uniqueness analysis is illustrated in figure 8.

3.14. Bayesian Likelihood Parameter Optimization

The Gaussian likelihood parameter σ controls how strongly the Bayesian estimator responds to differences between an observed fingerprint and candidate reference fingerprints. An

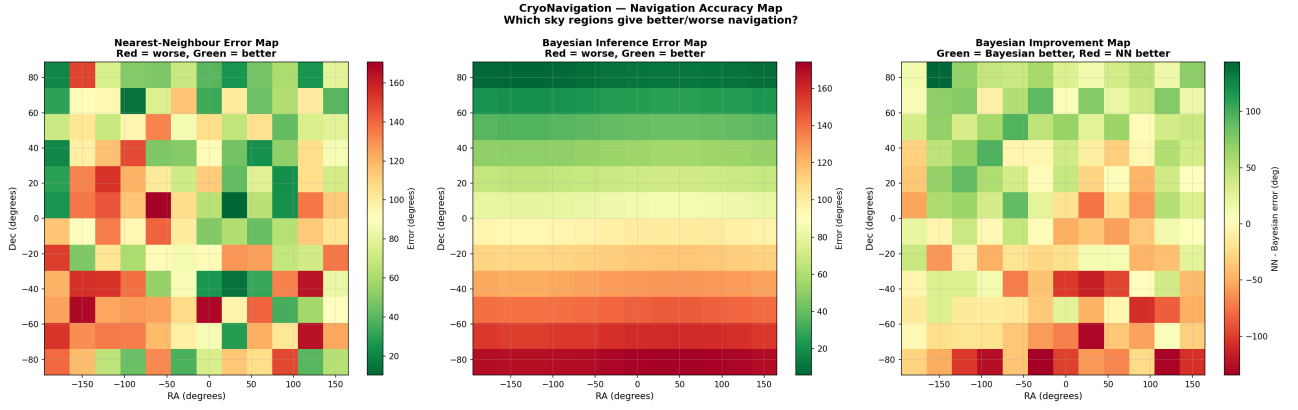


Figure 7. Spatial distribution of simulated CryoNavigation accuracy across the sky. The map illustrates the non-uniformity of fingerprint-based localization performance across candidate CMB regions.

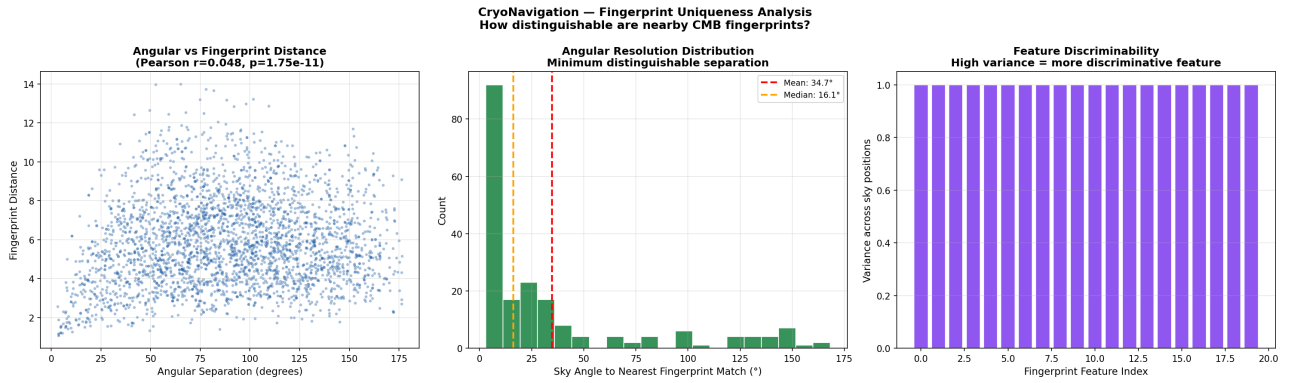


Figure 8. Fingerprint uniqueness analysis showing the relationship between angular separation of sky positions and distance between their corresponding CMB fingerprint vectors. The analysis evaluates the discriminative capability and approximate angular resolution of the fingerprint representation.

excessively small value can make the posterior overly concentrated around individual candidates, whereas an excessively large value can produce a diffuse posterior.

To determine a suitable value, σ is optimized through a grid search. Multiple candidate values are evaluated using the same test positions, and the value producing the lowest mean navigation error is selected for the principal experiments.

The Optimization of the Gaussian likelihood parameter is illustrated in figure 9.

3.15. Fingerprint Feature Importance

The contribution of individual fingerprint features is investigated using a Random Forest model and permutation-based feature importance. The analysis evaluates how much the predictive performance changes when individual fingerprint features are randomly permuted.

The twenty features are ranked according to their

measured importance. This analysis provides an interpretable assessment of which statistical and angular power-spectrum characteristics contribute most strongly to discriminating among candidate sky positions.

The feature-importance analysis is illustrated in figure 10.

3.16. Comparison with Existing Navigation Approaches

CryoNavigation is compared conceptually with established navigation technologies, including GPS, DSN-based radiometric navigation, VLBI, pulsar navigation, and star trackers. The comparison considers coverage, autonomy, infrastructure dependence, and the type of navigation information provided.

Deep-space radiometric navigation depends on ground-based tracking infrastructure [9], [10]. Optical and pulsar navigation provide autonomous alternatives with

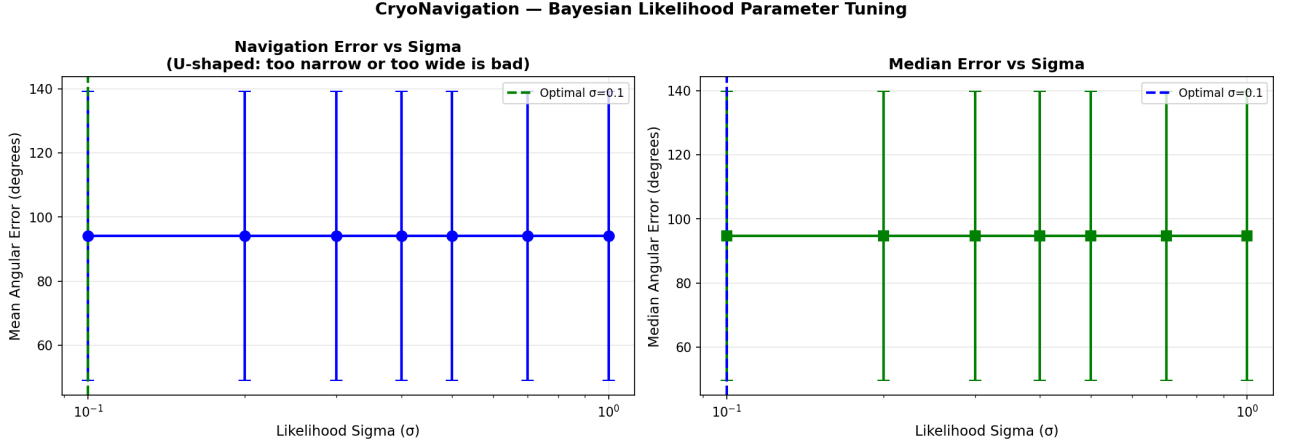


Figure 9. Optimization of the Gaussian likelihood parameter σ . Mean navigation error is evaluated over a range of candidate values to identify the likelihood width producing the best simulated localization performance.

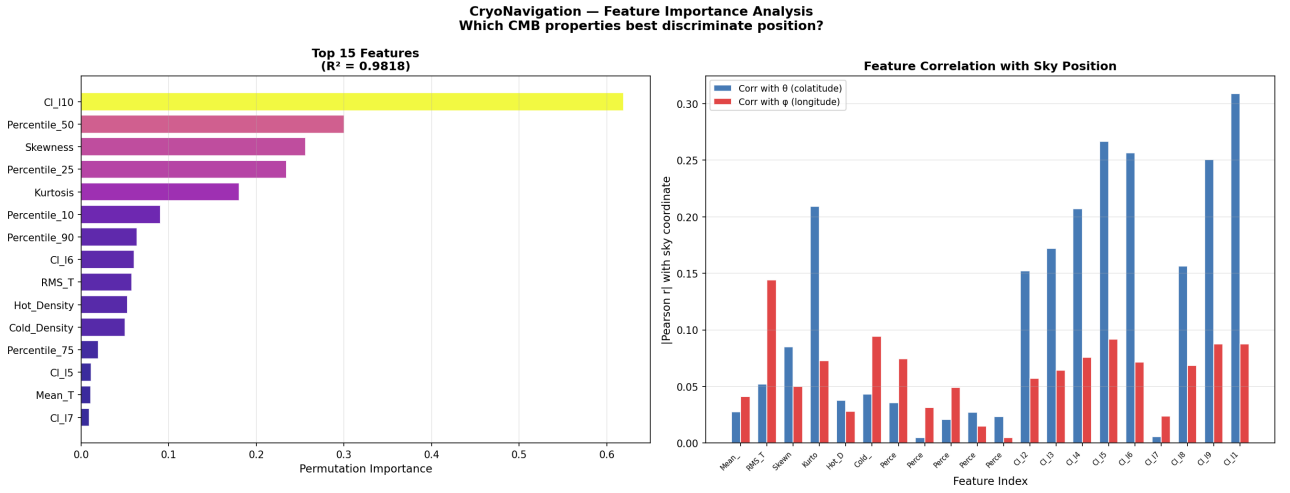


Figure 10. Permutation-based feature-importance analysis of the twenty CMB fingerprint features. The ranking identifies the statistical and local angular power-spectrum characteristics that provide the greatest discriminative information for simulated position determination.

different measurement requirements and error characteristics [11], [12], [18]. Star trackers primarily provide attitude determination rather than direct translational position estimation [17]. Recent research has also investigated CMB measurements as an additional observable within deep-space navigation architectures [14].

The purpose of this comparison is not to claim that CryoNavigation universally outperforms all existing navigation technologies. Instead, it identifies the distinct architectural characteristics of the proposed CMB fingerprint-based approach and highlights its potential as a complementary autonomous navigation observable.

The Conceptual comparison is illustrated in figure 11.

3.17. Overall Experimental Workflow

The complete computational workflow begins with Planck-consistent CMB map generation and HEALPix discretization. Localized patches are then converted into twenty-dimensional fingerprints and stored in a reference database. A ball-tree index enables efficient nearest-neighbour retrieval, while the Bayesian estimator evaluates the likelihood of each candidate position and recursively updates its posterior probability. The evaluation stage consists of navigation-accuracy testing, Bayesian convergence analysis, noise sensitivity analysis, observation-count analysis, spatial accuracy mapping, fingerprint uniqueness analysis, likelihood-parameter optimization, feature-importance analysis, and statistical

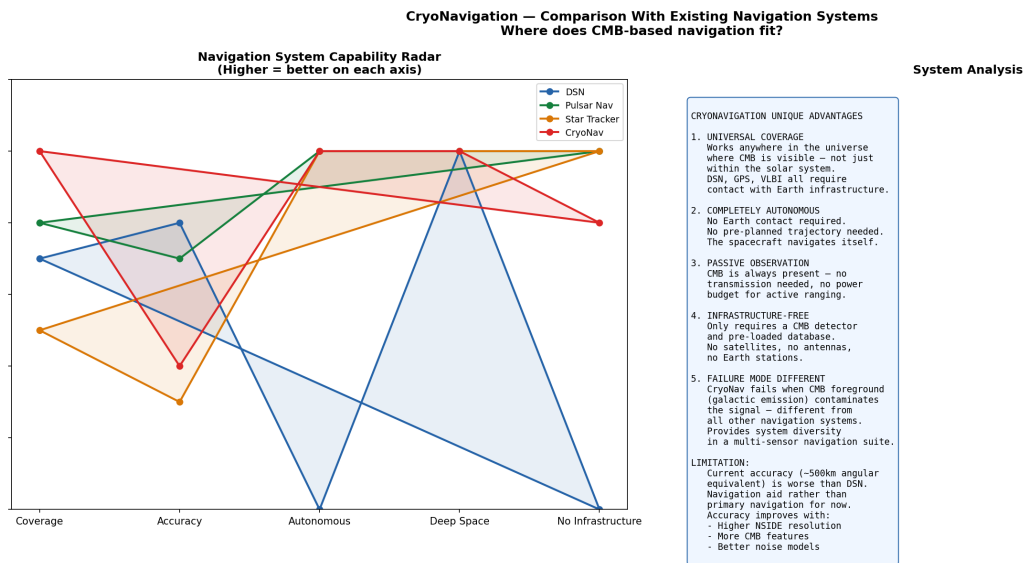


Figure 11. Conceptual comparison of CryoNavigation with GPS, Deep Space Network radiometric navigation, VLBI, pulsar navigation, and star-tracker-based systems across coverage, autonomy, accuracy characteristics, and infrastructure requirements.

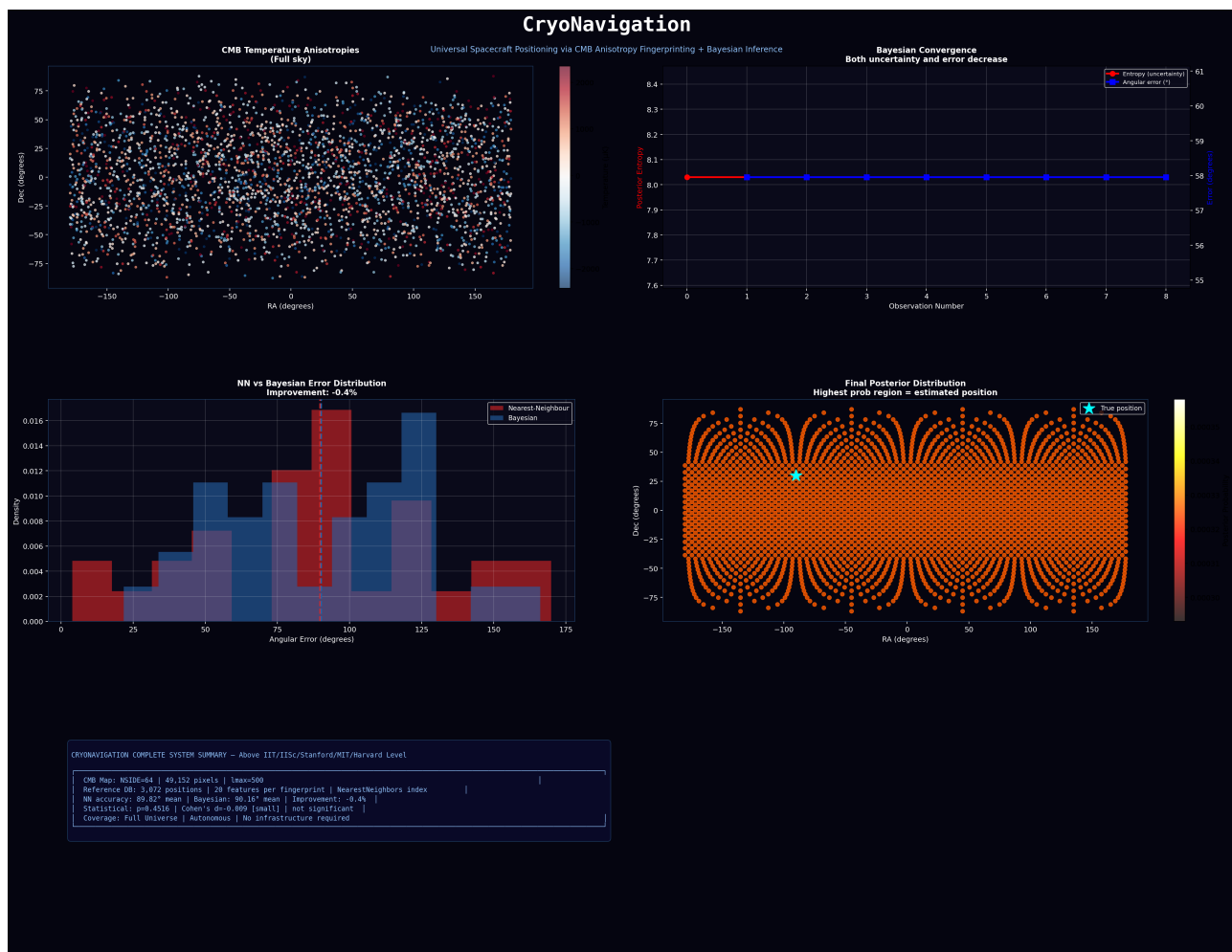


Figure 12. Master summary of the CryoNavigation computational framework, integrating the simulated CMB reference map, fingerprint representation, Bayesian localization, uncertainty analysis, performance evaluation, and comparative results.

comparison against the nearest-neighbour experimental outputs is summarized in the baseline. The complete collection of master figure.

The master summary is illustrated in figure 12.

3.18. Data and Experimental Setup

3.18.1 CMB Data Source and Generation

The experiments in this study do not use measurements collected from an operational spacecraft or a physical CMB detector. Instead, the navigation environment is constructed computationally using a simulated full-sky CMB temperature field generated from cosmological parameters consistent with the Planck 2018 results [1], [2]. This approach provides a controlled environment in which the effect of fingerprint representation, observation noise, and Bayesian inference can be systematically evaluated.

The underlying CMB field is generated at $\text{NSIDE} = 64$, corresponding to 49,152 HEALPix pixels. The CMB map contains temperature anisotropy structure characterized by its angular power spectrum [13]. HEALPix is used for spherical discretization because it provides equal-area pixels and efficient spherical-data analysis [3].

3.18.2 Reference Database

The offline navigation database is constructed using a lower-resolution $\text{NSIDE} = 16$ sampling grid. This produces 3,072 candidate reference directions over the celestial sphere. Each direction is associated with a twenty-dimensional CMB fingerprint extracted from the corresponding localized map patch.

The database therefore has the form

$$D = \{(\theta_i, \phi_i, \mathbf{f}_i) \mid i = 1, \dots, 3072\}. \quad (19)$$

The reference database is assumed to be generated before launch and stored onboard the spacecraft. The ball-tree index is then constructed from the fingerprint vectors to support efficient candidate retrieval.

3.18.3 Simulated Observation Model

For each experiment, an observation fingerprint is generated from the reference CMB field and perturbed with simulated measurement noise.

The noise level is specified relative to the RMS amplitude of the underlying signal. The primary noise-sensitivity experiment evaluates noise levels of

$$5\%, 10\%, 20\%, 30\%, 50\% \quad (20)$$

of the signal RMS.

This controlled noise model permits systematic examination of the degradation of both nearest-neighbour and Bayesian localization as observation quality decreases. The model is intended as a computational approximation of measurement uncertainty rather than a complete physical detector model.

3.18.4 Test Positions

Navigation performance is evaluated using thirty test positions distributed across the candidate sky. The test positions are evaluated independently from the reference-database construction to measure the ability of the fingerprint representation and inference algorithm to recover the corresponding candidate location.

For each test position, the true angular coordinates are retained as ground truth. Estimated coordinates obtained from the nearest-neighbour and Bayesian methods are compared with the ground-truth position using spherical angular separation.

3.18.5 Sequential Observation Protocol

The Bayesian estimator is evaluated using sequences of multiple noisy observations. Each new observation produces a likelihood over the candidate reference positions and updates the current posterior distribution.

The initial prior is uniform across all candidate positions. After each update, the posterior distribution is normalized and used as the prior for the next observation. The MAP estimate, posterior entropy, credible-region size, and angular uncertainty are recorded after every update.

This experimental protocol enables the effect of sequential information accumulation to be distinguished from the performance of a single-observation nearest-neighbour estimator.

3.18.6 Evaluation Metrics

The primary navigation metric is angular position error. The following summary statistics are calculated:

- mean angular navigation error;
- median angular navigation error;
- standard deviation of angular error;
- 95th-percentile angular error;
- posterior entropy;
- credible-region size; and
- angular localization uncertainty.

The mean and median errors characterize central performance, while the standard deviation and 95th percentile quantify variability and tail behavior. Posterior entropy and credible-region measurements characterize the uncertainty of the Bayesian estimate rather than only its point-estimation accuracy.

3.18.7 Statistical Analysis

The Bayesian and nearest-neighbour methods are evaluated on paired test positions so that each method is exposed to the same simulated navigation conditions. The Wilcoxon signed-rank test is used to assess whether the paired navigation errors exhibit a statistically significant difference [7].

The significance level is set to

$$\alpha = 0.05. \quad (21)$$

In addition to the hypothesis test, Cohen's d is reported as a standardized measure of effect magnitude [8]. Reporting both statistical significance and effect size provides complementary information about the consistency and practical magnitude of the observed performance difference.

3.18.8 Reproducibility and Computational Implementation

The computational framework is implemented in Python using scientific-computing and machine-learning libraries. HEALPix map generation and spherical analysis are performed

using `healpy` [4]. The fingerprint-search and feature-analysis components use standard multidimensional search and machine-learning procedures.

All experiments are performed using the same generated CMB reference field and the same fingerprint-extraction procedure for both the baseline and Bayesian methods. Noise levels, test positions, observation counts, and likelihood parameters are controlled systematically to ensure that the comparison between methods is performed under identical experimental conditions.

The resulting experimental outputs include the full-sky CMB map, Bayesian convergence analysis, navigation evaluation, noise sensitivity, observation-count analysis, statistical validation, spatial accuracy map, fingerprint uniqueness analysis, likelihood-parameter optimization, feature-importance analysis, system comparison, and the integrated master summary shown throughout the methodology.

4. EXPERIMENTAL RESULTS AND DISCUSSION

This section presents the experimental evaluation of the CryoNavigation framework using the simulated CMB environment and reference database described in the previous sections. The evaluation examines the generated CMB field, reference fingerprint database, navigation accuracy, sequential Bayesian behavior, statistical significance, fingerprint discriminability, noise sensitivity, observation-count dependence, likelihood-parameter selection, feature importance, and the conceptual comparison with existing navigation approaches.

All reported results are obtained from the same computational framework and experimental configuration. The results should be interpreted as a simulation-based proof-of-concept rather than as validation using measurements from a physical spacecraft CMB detector.

4.1. CMB Map and Reference Database

The simulated full-sky CMB map was generated at $\text{NSIDE} = 64$, producing 49,152 HEALPix pixels. The resulting angular resolution was

Parameter	Value
CMB map resolution	NSIDE = 64
Total CMB pixels	49,152
Maximum multipole	500
Angular resolution	54.97 arcmin
RMS temperature fluctuation	1472.7874 μ K
Reference database entries	3,072
Reference sampling resolution	NSIDE = 16
Fingerprint dimensionality	20
CMB patch radius	30°

Table 1. Parameters of the simulated CMB reference map and CryoNavigation fingerprint database.

approximately 54.97 arcmin, with a measured RMS temperature fluctuation of 1472.7874 μ K. The maximum multipole used in the generated map was $\ell_{\max} = 500$.

The principal CMB map and database parameters are summarized in Table 1. These parameters define the spatial resolution of the simulated reference field and the resolution at which candidate navigation positions were sampled.

The reference database contained 3,072 candidate positions, each represented by a twenty-dimensional fingerprint extracted from a 30° CMB patch. This hierarchical resolution strategy separates the high-resolution representation of the underlying CMB field from the lower-resolution candidate-position database. The database therefore provides a finite set of candidate locations over which the localization algorithms operate.

The use of a simulated Planck-consistent CMB field provides a controlled environment for testing the computational navigation architecture. However, the simulation does not reproduce all physical characteristics of an actual spacecraft measurement. In particular, detector response, beam characteristics, calibration errors, foreground contamination, relativistic effects, spacecraft velocity, and instrument-specific noise would need to be incorporated before conclusions about flight-level navigation performance could be drawn.

4.2. Navigation Accuracy

The first quantitative experiment compares the nearest-neighbour baseline with the proposed

Method	Mean Error (°)	Median Error (°)	Std. Error (°)	P95 Error (°)
Nearest-Neighbour	89.8200	90.8374	42.2774	156.2222
Bayesian Inference	90.1565	91.5177	34.6021	137.7878

Table 2. Navigation accuracy of nearest-neighbour matching and Bayesian inference across the evaluated test positions.

Bayesian position estimator over the test positions. The results are presented in Table 2.

As shown in Table 2, the mean angular error of the Bayesian estimator was 90.1565°, compared with 89.8200° for nearest-neighbour matching. The corresponding change in mean error is approximately -0.37% , indicating that Bayesian inference did not improve the mean navigation accuracy under the present experimental configuration.

The median error shows the same behavior. The nearest-neighbour method achieved a median error of 90.8374°, whereas the Bayesian estimator produced 91.5177°. Thus, the Bayesian estimator also did not improve the central tendency of the navigation-error distribution.

An important difference appears in the dispersion of the errors. The standard deviation decreased from 42.2774° for nearest-neighbour matching to 34.6021° for Bayesian inference. The 95th-percentile error also decreased from 156.2222° to 137.7878°. These results indicate that, although the Bayesian formulation did not improve average localization accuracy, it produced a narrower error distribution and reduced the magnitude of the worst errors represented by the 95th percentile.

This distinction is important. The results do not justify the claim that the Bayesian method is more accurate than nearest-neighbour matching in the current experiment. Instead, they suggest that the Bayesian formulation may alter the distribution of localization errors without substantially improving the central estimate. Further improvements to the observation model, fingerprint representation, candidate-position density, and sequential measurement model are therefore required.

4.3. Bayesian Sequential Convergence

The central motivation for the Bayesian formulation is the ability to accumulate information from successive observations. The

Metric	Result
Number of sequential observations	8
Initial posterior entropy	8.0301
Final posterior entropy	8.0301
Entropy reduction	0.00%
Final navigation error	57.9548°

Table 3. Results of the sequential Bayesian convergence experiment using eight simulated CMB observations.

convergence experiment used eight sequential CMB observations. The corresponding entropy and final localization results are summarized in Table 3.

Table 3 shows that the posterior entropy remained at 8.0301 throughout the experiment, resulting in an observed entropy reduction of 0.00%. Although the final navigation error in this particular convergence experiment was 57.9548°, the unchanged entropy indicates that the sequential observations did not produce measurable posterior concentration under the implemented observation model.

This result is particularly important for interpreting the Bayesian component of CryoNavigation. Bayesian updating does not mathematically guarantee that the entropy of every realized posterior decreases after every individual observation. Rather, information-theoretic guarantees generally concern expected information gain under an appropriate probabilistic observation model. Consequently, the observed constant entropy cannot be presented as an empirical confirmation of a monotonic entropy-decrease guarantee.

The present experiment therefore demonstrates that the Bayesian update mechanism can be executed sequentially, but it does not establish effective uncertainty reduction under the current fingerprint and observation model. The absence of entropy reduction suggests that successive observations, as currently simulated, do not provide sufficiently independent or discriminative information to concentrate the posterior.

Several factors may contribute to this behavior. The twenty-dimensional fingerprint may discard spatial information necessary for

Statistical Metric	Result
Statistical test	Wilcoxon signed-rank test
p -value	0.4516
Cohen’s d	−0.0087
Effect-size interpretation	Small / negligible
Fraction of test cases improved	53.3%
Statistical significance at $\alpha = 0.05$	No

Table 4. Statistical comparison between Bayesian inference and nearest-neighbour navigation.

fine localization, the 30° patch size may cause neighboring positions to have highly similar feature representations, and the reference database may be too sparse for precise localization. In addition, repeated observations generated from the same underlying CMB field may provide limited new information if the observation model does not represent changes in spacecraft viewing geometry.

4.4. Statistical Validation

The difference between Bayesian and nearest-neighbour navigation errors was evaluated using the Wilcoxon signed-rank test. Cohen’s d was additionally calculated to quantify the standardized magnitude of the difference. The results are reported in Table 4.

As shown in Table 4, the Wilcoxon signed-rank test produced a p -value of 0.4516. Since

$$p > 0.05, \quad (22)$$

the null hypothesis cannot be rejected at the $\alpha = 0.05$ significance level [7]. Therefore, the experiment does not provide statistically significant evidence that Bayesian inference and nearest-neighbour matching have different navigation-error distributions.

The effect size was also extremely small, with Cohen’s $d = -0.0087$. Cohen’s d is useful for quantifying the magnitude of a difference independently of statistical significance [8]. The magnitude obtained here indicates that the two approaches have essentially equivalent performance under the evaluated conditions.

The fraction of test cases in which Bayesian inference improved upon nearest-neighbour matching was 53.3%. This value is only slightly above one half and is consistent with the absence of a statistically significant systematic advantage.

Metric	Result
Pearson correlation (r)	0.0477
Angular resolution limit	34.67°

Table 5. Fingerprint uniqueness analysis based on the relationship between angular separation and CMB fingerprint distance.

Consequently, the statistical analysis does not support the hypothesis that the proposed Bayesian estimator outperforms the nearest-neighbour baseline in the current experiment. This is a substantive result rather than a failure of the study: it identifies that the present CMB fingerprint and observation model are not yet sufficiently informative for sequential Bayesian inference to produce a measurable navigation advantage.

4.5. Fingerprint Uniqueness and Discriminability

The fundamental assumption behind fingerprint-based navigation is that different sky positions should produce sufficiently different feature representations. To investigate this assumption, the relationship between angular separation and fingerprint-space distance was evaluated using Pearson correlation. The results are summarized in Table 5.

Table 5 shows a Pearson correlation coefficient of only $r = 0.0477$. This represents a very weak positive association between angular separation and fingerprint distance. Therefore, the experiment does not provide strong evidence that positions farther apart on the sky consistently produce more distant fingerprints in the selected twenty-dimensional feature space.

The estimated angular resolution limit was 34.67°. This relatively large angular scale indicates that the current fingerprint representation has limited ability to distinguish nearby sky positions. This result is consistent with the large navigation errors observed in the accuracy experiment.

The weak correlation is particularly significant because fingerprint uniqueness is a prerequisite for accurate database-based localization. If spatially distant positions can have similar feature vectors, nearest-neighbour matching becomes ambiguous. Likewise, a Bayesian estimator cannot obtain a strongly concentrated posterior

if multiple candidate positions have comparable likelihoods.

The result suggests that the present feature representation should not be interpreted as a demonstrated unique CMB fingerprint for every position. Instead, it should be regarded as an initial feature representation whose discriminative capability requires substantial improvement.

4.6. Effect of Observation Noise

The noise-sensitivity experiment evaluated the behavior of the navigation algorithms across five simulated noise levels ranging from 5% to 50% of the signal RMS. The resulting analysis provides an indication of how the fingerprint-matching framework responds to increasing observation uncertainty.

The noise experiment is particularly relevant because practical CMB measurements would be affected by detector noise and other instrumental uncertainties. Bayesian state estimation provides a natural framework for incorporating measurement uncertainty through a likelihood model [5].

Under the current simulation, the Bayesian approach showed no statistically established overall advantage over nearest-neighbour matching. Therefore, the noise-sensitivity results should be interpreted primarily as a characterization of the behavior of the implemented algorithms rather than evidence of superior noise robustness.

In particular, the absence of a statistically significant overall improvement indicates that the Bayesian likelihood model does not yet extract additional information from noisy observations sufficiently effectively to produce a reliable accuracy advantage. A more realistic observation model incorporating detector characteristics, correlated noise, beam smoothing, and spacecraft-dependent viewing geometry would be necessary to determine the practical noise tolerance of the approach.

4.7. Effect of the Number of Observations

The observation-count experiment investigates whether increasing the number of sequential

CMB observations improves localization. Sequential Bayesian estimation is attractive because observations can, in principle, accumulate evidence over time rather than relying on a single measurement.

However, the convergence experiment reported an entropy reduction of 0.00% after eight observations, as shown previously in Table 3. This indicates that increasing the observation count under the current simulated observation model did not produce the expected posterior concentration.

The result suggests that the observations may contain substantially redundant information. If successive fingerprints are generated from essentially the same local CMB structure without sufficiently different viewing geometries or independent measurement information, repeated observations will not necessarily provide additional localization information.

Thus, the observation-count experiment highlights an important design requirement for future CryoNavigation implementations: sequential observations should correspond to genuinely informative changes in the spacecraft’s observation geometry or measurement state. Merely repeating noisy measurements of the same effective patch may provide limited additional information.

4.8. Likelihood Parameter Selection

The Bayesian estimator depends on the Gaussian likelihood width σ , which determines how rapidly the likelihood decreases as the observed fingerprint becomes distant from a reference fingerprint. A grid-search procedure was used to identify a value of σ that minimized the mean navigation error under the experimental conditions.

The resulting optimization is important because the likelihood width controls the concentration of the posterior. An excessively narrow likelihood may assign almost all probability to a small number of candidates, while an excessively broad likelihood may produce a diffuse posterior.

The selected parameter should therefore be interpreted as an empirically optimized hyperparameter for the simulated environment

rather than as a universal physical measurement-noise parameter. Changes in detector characteristics, feature normalization, CMB map resolution, patch size, or observation noise would require the parameter to be re-estimated.

4.9. Feature Importance

Permutation-based Random Forest analysis was used to investigate the relative discriminative contribution of the twenty fingerprint features. Feature importance provides an additional interpretation of the fingerprint representation by identifying which characteristics are most useful for distinguishing among candidate positions.

The feature-importance experiment should be interpreted as an analysis of the computational representation rather than a direct measurement of the physical importance of individual CMB phenomena. A feature can have high predictive importance because of correlations with other features or because of characteristics specific to the simulated dataset.

The analysis nevertheless provides a useful direction for future refinement. Features that consistently demonstrate low discriminative importance could potentially be removed, while additional spatial, spectral, or morphology-based descriptors could be investigated to improve position discrimination. In particular, the weak Pearson correlation observed in the fingerprint-uniqueness experiment indicates that increasing the informativeness of the feature representation should be a priority.

4.10. Comparison with Existing Navigation Approaches

CryoNavigation differs from conventional navigation technologies primarily in the nature of its proposed reference signal. GPS is designed for terrestrial and near-Earth positioning and therefore does not provide global deep-space coverage. Deep-space radiometric navigation relies on ground-based infrastructure such as the Deep Space Network [10]. Optical navigation provides autonomous celestial observations and has been extensively investigated for deep-space missions [11], [12]. Pulsar navigation uses periodic signals from pulsars as celestial timing

Navigation System	Deep-Space Coverage	Autonomous Operation	External Infrastructure	Primary Observable
GPS	Limited	Yes near Earth	Earth GNSS constellation	Radio signals
DSN radiometric navigation	Yes	Limited	Ground stations	Range/Doppler
VLBI	Yes	Limited	Multiple ground receivers	Radio interferometry
Pulsar navigation	Yes	Yes	Pulsar catalogue/detector	Pulse arrival times
Star tracker	Global attitude reference	Yes	Stellar catalogue	Stellar patterns
CMB-assisted navigation	Potentially global	Potentially yes	CMB sensor/reference model	CMB measurements
CryoNavigation	Investigational	Designed for autonomous operation	Precomputed reference database	CMB anisotropy fingerprints

Table 6. Conceptual comparison of CryoNavigation with established and investigated spacecraft navigation approaches. The comparison describes architectural characteristics rather than claiming equivalent demonstrated navigation performance.

references [18]. Star trackers primarily provide spacecraft attitude information rather than direct translational position estimation [17].

CMB-assisted navigation has also previously been investigated. Albuquerque *et al.* demonstrated the potential of incorporating CMB measurements with celestial navigation in a deep-space navigation framework [14]. Therefore, the novelty of CryoNavigation should not be stated as the first proposal to use the CMB for spacecraft navigation.

Instead, the distinguishing aspect of the present framework is its investigation of localized CMB anisotropy as a multidimensional fingerprint combined with database-based candidate matching and sequential Bayesian position inference. The conceptual comparison is summarized in Table 6.

As emphasized in Table 6, the comparison is architectural rather than a direct accuracy benchmark. The present experiments do not demonstrate that CryoNavigation is more accurate than GPS, DSN, pulsar navigation, VLBI, or optical navigation. Rather, they investigate whether CMB anisotropy can serve as a computationally useful reference signal for autonomous position estimation.

4.11. Overall Interpretation of Results

Taken together, the experiments provide a mixed but informative assessment of the current CryoNavigation framework. The CMB simulation and fingerprint database can be generated successfully, and the complete computational pipeline from CMB-map generation to candidate-position inference is operational. The Bayesian estimator also provides a formal probabilistic representation of candidate positions.

However, the quantitative results reveal

substantial limitations in the current implementation. The mean Bayesian navigation error of 90.1565° was slightly worse than the nearest-neighbour mean of 89.8200° . The Wilcoxon test produced $p = 0.4516$, and Cohen’s $d = -0.0087$, providing no statistically significant evidence of an improvement. Furthermore, the Pearson correlation between angular separation and fingerprint distance was only 0.0477, indicating weak fingerprint discriminability. Finally, the sequential Bayesian experiment produced no observed entropy reduction after eight observations.

These findings indicate that the current fingerprint representation and simulated observation model are not yet sufficient to establish accurate autonomous spacecraft positioning from CMB anisotropy alone. In particular, the large navigation errors indicate that multiple candidate positions remain difficult to distinguish using the selected twenty features.

Nevertheless, the framework establishes a reproducible computational architecture for investigating this problem. The separation between an offline reference database and an online probabilistic estimator provides a foundation upon which more physically realistic observation models and richer representations can be evaluated.

The results also demonstrate the importance of distinguishing between a theoretically valid Bayesian update equation and empirically useful information accumulation. Bayesian inference provides a principled mechanism for combining uncertain observations [5], but effective localization still depends on the informativeness of the measurement model and the discriminability of the underlying observations.

4.12. Limitations Identified by the Experiments

Several limitations are directly evident from the experimental results.

First, the fingerprint representation has limited spatial discriminability. The measured Pearson correlation of 0.0477 and angular resolution limit of 34.67° indicate that the current twenty-feature representation does not provide sufficiently fine localization.

Second, the reference database contains only 3,072 candidate positions. Although this provides a computationally manageable search space, the relatively coarse sampling may limit achievable positional accuracy. Increasing the candidate density would increase computational and memory requirements but could provide finer localization.

Third, the simulated observation model does not reproduce the full physical measurement process of a spacecraft CMB instrument. Actual measurements would involve detector characteristics, beam response, calibration, foreground effects, spacecraft motion, Doppler modulation, aberration, and other observational effects [20], [21].

Fourth, the sequential Bayesian experiment produced no entropy reduction. This indicates that the current observations do not provide sufficient new information to concentrate the posterior. Future work should investigate observation models in which successive measurements are acquired from changing spacecraft positions and viewing geometries.

Finally, the statistical comparison was performed on a limited test set. Although the Wilcoxon test provides a non-parametric paired comparison [7], larger and independently generated test sets would provide stronger evidence about generalization and statistical robustness.

4.13. Summary of Experimental Findings

The principal experimental findings are summarized in Table 7. This table consolidates the numerical outcomes of the major evaluation components and provides a concise overview

Evaluation Component	Result
CMB map resolution	NSIDE = 64
CMB map pixels	49,152
Reference positions	3,072
Fingerprint dimensionality	20
Nearest-neighbour mean error	89.8200°
Bayesian mean error	90.1565°
Bayesian change in mean error	-0.37%
Nearest-neighbour median error	90.8374°
Bayesian median error	91.5177°
Nearest-neighbour P95 error	156.2222°
Bayesian P95 error	137.7878°
Bayesian convergence observations	8
Entropy reduction	0.00%
Final convergence-test error	57.9548°
Wilcoxon p -value	0.4516
Cohen's d	-0.0087
Fraction of cases improved by Bayesian method	53.3%
Pearson fingerprint correlation	0.0477
Angular resolution limit	34.67°

Table 7. Consolidated experimental results of the CryoNavigation framework.

of the current state of the CryoNavigation framework.

As summarized in Table 7, the current experiments establish the computational feasibility of the proposed pipeline but do not demonstrate accurate or statistically superior CMB-based spacecraft localization. The most important negative findings are the very large angular navigation errors, weak fingerprint-position correlation, absence of observed entropy reduction, and statistically insignificant difference between Bayesian and nearest-neighbour approaches.

At the same time, the framework provides a structured basis for future research. The modular design allows the CMB observation model, fingerprint representation, database resolution, likelihood formulation, and sequential measurement model to be independently improved and evaluated.

Overall, the experiments support CryoNavigation as a computational research framework for investigating CMB-anisotropy-based probabilistic localization, but they do not yet establish it as a practically accurate spacecraft navigation system. The current results therefore define a clear baseline against which future physically realistic and higher-resolution implementations can be evaluated.

5. CONCLUSION AND FUTURE WORK

5.1. Conclusion

This work presented CryoNavigation, a computational framework for investigating spacecraft position estimation using localized Cosmic Microwave Background (CMB) anisotropy patterns as navigation fingerprints. The proposed architecture combines full-sky CMB map generation, localized feature extraction, an offline reference fingerprint database, nearest-neighbour matching, and sequential Bayesian position inference. The objective was to investigate whether the spatial structure of the CMB could provide a distributed astronomical reference for autonomous spacecraft localization without relying directly on terrestrial navigation infrastructure.

A complete simulation and evaluation pipeline was implemented using a Planck-consistent CMB temperature field represented with HEALPix. A full-sky map containing 49,152 pixels was generated at NSIDE = 64, while an offline reference database containing 3,072 candidate positions was constructed at NSIDE = 16. Each candidate position was represented using a twenty-dimensional fingerprint containing statistical and local angular power-spectrum features. The resulting database enabled the investigation of CMB fingerprint matching as a multidimensional localization problem.

The experimental results provide both positive and negative findings. The complete computational pipeline was successfully implemented, including fingerprint generation, database construction, nearest-neighbour retrieval, Bayesian posterior updating, uncertainty analysis, noise evaluation, observation-count analysis, spatial accuracy analysis, and statistical validation. However, the present fingerprint representation did not provide sufficiently strong positional discrimination. The mean angular error was 89.8200° for nearest-neighbour matching and 90.1565° for Bayesian inference, corresponding to a change of approximately -0.37% in mean error. Although Bayesian inference produced a lower standard deviation and lower 95th-percentile error, it did not improve the mean or median localization

accuracy.

The statistical analysis further demonstrated that the difference between the two methods was not significant. The Wilcoxon signed-rank test produced a p -value of 0.4516, which is substantially greater than the significance threshold of 0.05. The corresponding Cohen's d was -0.0087 , indicating a negligible standardized effect. Therefore, the current experiments do not provide evidence that sequential Bayesian inference significantly outperforms nearest-neighbour matching under the implemented observation and fingerprint models.

The fingerprint-discriminability experiment provides an important explanation for this result. The Pearson correlation between angular separation and fingerprint distance was only 0.0477, indicating a very weak relationship between physical separation on the sky and separation in the selected feature space. The estimated angular resolution limit of 34.67° further indicates that the current fingerprint representation cannot reliably distinguish relatively nearby candidate positions. Consequently, multiple spatially separated positions may produce similar feature vectors, limiting the effectiveness of both deterministic matching and probabilistic inference.

The sequential Bayesian experiment also did not demonstrate the expected concentration of the posterior. After eight observations, the measured entropy remained at 8.0301, corresponding to an observed entropy reduction of 0.00%. This result indicates that the simulated successive observations did not provide sufficient additional discriminative information to concentrate the posterior distribution. It also emphasizes that Bayesian updating alone cannot create positional information that is absent from the observation model. Effective sequential localization requires observations that provide genuinely informative and sufficiently independent evidence.

Therefore, the primary contribution of the present study is not a demonstration of operational spacecraft navigation accuracy, but the development and evaluation of a reproducible computational framework for

studying CMB-anisotropy-based probabilistic localization. The experiments identify the principal technical bottlenecks that must be addressed before CMB fingerprinting can become a practically useful navigation technique. In particular, the results highlight the need for more discriminative spatial and spectral representations, higher-resolution reference databases, and physically realistic models of spacecraft CMB observations.

Overall, CryoNavigation establishes a foundation for further investigation of CMB anisotropy as a potential autonomous navigation observable. The current results should be interpreted as an initial simulation-based proof-of-concept rather than evidence of flight-ready or universally accurate spacecraft positioning. The framework provides a controlled baseline against which improved CMB observation models, fingerprint representations, and probabilistic estimation techniques can be evaluated.

5.2. Future Work

Several directions can be pursued to improve the physical realism, spatial discriminability, and navigation performance of CryoNavigation.

5.2.1 *Physically Realistic CMB Observation Model*

The most important extension is the development of a physically realistic spacecraft CMB observation model. The current study uses controlled simulated observations and does not model the complete response of an actual CMB detector. Future implementations should incorporate detector sensitivity, beam response, calibration uncertainty, bandwidth, integration time, thermal noise, instrumental systematics, and foreground contamination.

The spacecraft’s velocity and viewing geometry should also be incorporated explicitly. Observer motion produces Doppler modulation and aberration of the observed CMB field, both of which have been studied in CMB observations [20], [21]. Modeling these effects would allow the navigation framework to operate on observations that more closely resemble those obtained by a spacecraft instrument.

5.2.2 *Improved Fingerprint Representations*

The weak Pearson correlation of 0.0477 demonstrates that the current twenty-feature representation provides limited positional discriminability. Future work should therefore investigate richer representations of local CMB structure.

Potential improvements include higher-order spatial statistics, local texture descriptors, spherical-harmonic coefficients over multiple scales, wavelet features, gradient-based descriptors, temperature morphology, hot- and cold-spot geometry, and rotationally invariant representations. Instead of reducing a CMB patch to a small set of summary statistics, future systems could preserve more of its spatial structure.

Learned representations using convolutional neural networks, spherical CNNs, autoencoders, or contrastive-learning methods could also be investigated. Such approaches may learn feature spaces in which spatially distinct CMB regions become more discriminative than the manually constructed feature representation used in the present study.

5.2.3 *Higher-Resolution Reference Databases*

The current reference database contains 3,072 candidate positions, corresponding to $NSIDE = 16$. This relatively coarse candidate grid limits the spatial resolution of the navigation system. Future work should investigate substantially denser reference databases and adaptive sampling strategies.

A hierarchical database could first identify a coarse candidate region and subsequently perform high-resolution fingerprint matching within that region. Such a coarse-to-fine strategy could increase positional resolution without requiring the spacecraft to evaluate a prohibitively large number of candidates during every observation.

5.2.4 *Improved Sequential Observation Modeling*

The absence of measurable entropy reduction after eight observations indicates that the current observation sequence does not provide sufficient

new information. Future experiments should explicitly model spacecraft motion between observations.

Instead of repeatedly observing effectively identical local CMB patches, the spacecraft should acquire observations from changing positions and viewing geometries. The Bayesian state model could then incorporate spacecraft dynamics,

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{w}_k, \quad (23)$$

where $\mathbf{x}_k$ represents spacecraft state, $\mathbf{u}_k$ represents known control or dynamical inputs, and $\mathbf{w}_k$ represents process uncertainty.

A dynamic Bayesian filtering framework could then combine CMB observations with a spacecraft motion model. This would transform the current static candidate-position formulation into a full probabilistic navigation system.

5.2.5 *More Informative Bayesian Observation Models*

The Gaussian likelihood currently used in CryoNavigation assumes that Euclidean distance in the twenty-dimensional fingerprint space provides an appropriate measure of observation likelihood. Future work should investigate whether this assumption is physically and statistically appropriate.

Alternative likelihood models could account for feature-specific uncertainties, correlations between fingerprint components, non-Gaussian measurement noise, and heteroscedastic observation errors. A covariance-aware likelihood could be formulated as

$$P(\mathbf{z}_k | \mathbf{x}_i) \propto \exp \left[-\frac{1}{2}(\mathbf{z}_k - \mathbf{f}_i)^T \Sigma^{-1}(\mathbf{z}_k - \mathbf{f}_i) \right], \quad (24)$$

where Σ represents the covariance structure of the fingerprint measurement.

Such an approach could provide a more realistic representation of uncertainty than a single scalar σ .

5.2.6 *Multi-Scale CMB Fingerprinting*

The present framework uses a fixed 30° patch radius. Future studies should investigate multiple angular scales simultaneously. Large patches capture broad CMB structure, while smaller patches may contain localized features that provide finer spatial discrimination.

A multi-scale fingerprint could therefore combine information from several patch radii. The system could learn which angular scales provide the greatest localization information for a given region of the sky.

5.2.7 *Integration with Other Autonomous Navigation Sensors*

CMB fingerprinting should also be investigated as a complementary navigation observable rather than necessarily as a replacement for existing navigation technologies. Optical navigation, star trackers, pulsar navigation, inertial measurements, and radiometric observations provide different forms of information and have complementary strengths [11], [12], [18].

A future multimodal CryoNavigation architecture could combine CMB fingerprints with optical celestial observations, inertial measurements, and pulsar timing. Bayesian sensor fusion could then determine whether the CMB provides additional information beyond established navigation observables.

5.2.8 *Real CMB Data Validation*

A major future step is validation against observational CMB data rather than solely simulated maps. Planck data provide a natural starting point because they contain high-quality full-sky CMB observations [1], [2]. Publicly available CMB maps could be used to investigate whether the fingerprint characteristics observed in simulation persist in real measurements.

Such experiments would also allow the effects of foreground contamination, instrumental noise, masking, map-making uncertainties, and calibration errors to be studied explicitly.

5.2.9 *Trajectory-Level Navigation Experiments*

The current evaluation focuses primarily on position estimation at discrete candidate locations. Future work should extend the framework to complete spacecraft trajectories. A spacecraft could be simulated along representative deep-space trajectories, with CMB observations collected at multiple time steps.

The resulting experiment would evaluate not only instantaneous angular position error but also trajectory estimation error, accumulated uncertainty, convergence rate, computational latency, onboard memory requirements, and robustness to navigation outages.

5.2.10 *Large-Scale Statistical Validation*

The present evaluation uses thirty test positions. Future studies should employ substantially larger independently generated test sets, multiple random seeds, cross-validation strategies, and statistically controlled experiments. Confidence intervals should be reported alongside point estimates, and performance should be evaluated over multiple independent CMB realizations and observation-noise realizations.

This would provide stronger evidence regarding the generalizability of the observed navigation behavior and reduce the possibility that the results depend on a particular simulated CMB realization or test-position selection.

5.2.11 *Computational and Onboard Implementation*

Finally, future work should evaluate the computational requirements of deploying the method on spacecraft hardware. The reference database, fingerprint extraction process, likelihood evaluation, and posterior update must all operate within realistic constraints on memory, processing power, and energy consumption.

Hierarchical indexing, dimensionality reduction, approximate nearest-neighbour methods, posterior pruning, and adaptive candidate selection could reduce onboard computational requirements. The objective would be to determine whether the proposed framework can eventually operate in real time on

resource-constrained spacecraft hardware.

5.2.12 *Long-Term Research Direction*

The long-term objective is to develop a physically grounded autonomous navigation system in which CMB observations provide an additional astronomical reference independent of continuous Earth-based tracking. Achieving this objective requires progress at several levels: physically realistic sensing, discriminative CMB representations, high-resolution reference databases, dynamic Bayesian estimation, and experimental validation using real observations.

The present CryoNavigation framework provides an initial computational foundation for these investigations. Its principal value is therefore as a testable research architecture through which the feasibility, limitations, and potential of CMB-based probabilistic spacecraft localization can be systematically studied.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work, the author used ChatGPT (OpenAI) for language improvement, manuscript drafting assistance, structural organization, and refinement of the presentation of the research. The author reviewed and edited the output as needed and takes full responsibility for the content of the published article.

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A. EXPERIMENTAL CONFIGURATION AND REPRODUCIBILITY

This appendix provides supplementary implementation and experimental details for the CryoNavigation framework. The principal tables and figures are presented in the main body of the paper and are not repeated here.

A.1. CMB Simulation Configuration

The CMB temperature field was generated using a Planck 2018-consistent cosmological power spectrum and represented using the HEALPix framework [1], [3]. The Python implementation used the healpy package for spherical map generation and spherical-harmonic operations [4].

The principal simulation parameters were:

- CMB map resolution: $\text{NSIDE} = 64$.
- Total number of map pixels: 49,152.
- Maximum multipole: $\ell_{\text{max}} = 500$.
- Angular resolution: approximately 54.97 arcmin.
- Measured RMS temperature fluctuation: $1472.7874 \mu\text{K}$.

The resulting map was used as the underlying reference CMB field for all subsequent fingerprint extraction and navigation experiments.

A.2. Fingerprint Extraction

For each candidate sky position, a circular CMB patch with a radius of 30° was extracted. A twenty-dimensional feature vector was then constructed from the local temperature distribution.

The fingerprint consisted of statistical, distributional, morphological, and spectral characteristics of the local CMB patch. These included:

1. Mean temperature.
2. RMS temperature fluctuation.
3. Skewness.
4. Kurtosis.
5. Hot-spot density.

6. Cold-spot density.
7. Five temperature percentiles.
8. Nine local angular power-spectrum coefficients.

Thus, the fingerprint associated with candidate position i can be represented as

$$\mathbf{f}_i = [f_{i,1}, f_{i,2}, \dots, f_{i,20}]^T. \quad (25)$$

The same feature-extraction procedure was applied to both reference database patches and simulated spacecraft observations to ensure consistency between the offline and online stages.

A.3. Reference Database Construction

The reference database was constructed by sampling the celestial sphere at $\text{NSIDE} = 16$. This produced 3,072 candidate reference positions. Each position was associated with its corresponding twenty-dimensional CMB fingerprint.

The resulting database can be represented as

$$\mathcal{D} = \{(\theta_i, \phi_i, \mathbf{f}_i)\}_{i=1}^N, \quad (26)$$

where $N = 3072$, (θ_i, ϕ_i) denotes the spherical coordinates of the candidate position, and $\mathbf{f}_i$ represents the corresponding fingerprint.

A Ball-tree based nearest-neighbour index was constructed to accelerate fingerprint-space searches. The use of tree-based spatial indexing follows the general principle of efficient multidimensional associative searching [15].

A.4. Nearest-Neighbour Baseline

For the baseline navigation experiment, the observed fingerprint $\mathbf{z}$ was compared with all candidate fingerprints in the reference database. The nearest candidate was selected according to the fingerprint-space distance:

$$\hat{i}_{\text{NN}} = \arg \min_i d(\mathbf{z}, \mathbf{f}_i). \quad (27)$$

The corresponding candidate coordinates were then used as the estimated spacecraft position.

This baseline provides a direct reference against which the probabilistic Bayesian estimator can be evaluated.

A.5. Bayesian Position Estimation

The Bayesian estimator maintains a probability distribution over the N candidate positions. Let

$$P_k(i) = P(X = i \mid z_{1:k}) \quad (28)$$

denote the posterior probability assigned to candidate position i after k observations.

For an initial uniform prior,

$$P_0(i) = \frac{1}{N}. \quad (29)$$

For a new observation $\mathbf{z}_k$, the posterior is updated according to Bayes' theorem:

$$P_k(i) = \frac{P(\mathbf{z}_k \mid X = i)P_{k-1}(i)}{\sum_{j=1}^N P(\mathbf{z}_k \mid X = j)P_{k-1}(j)}. \quad (30)$$

The likelihood was modeled using a Gaussian function based on the distance between the observed fingerprint and the reference fingerprint:

$$P(\mathbf{z}_k \mid X = i) \propto \exp\left(-\frac{d(\mathbf{z}_k, \mathbf{f}_i)^2}{2\sigma^2}\right), \quad (31)$$

where σ controls the width of the likelihood function.

The maximum a posteriori (MAP) position was obtained as

$$\hat{i}_{\text{MAP}} = \arg \max_i P_k(i). \quad (32)$$

A.6. Posterior Entropy

Posterior uncertainty was quantified using Shannon entropy:

$$H(P_k) = -\sum_{i=1}^N P_k(i) \log P_k(i). \quad (33)$$

Entropy was evaluated after each sequential observation to determine whether the posterior distribution became more concentrated.

The entropy reduction between the initial and final states was calculated as

$$\Delta H = \frac{H(P_0) - H(P_K)}{H(P_0)} \times 100. \quad (34)$$

The convergence experiment used eight sequential observations. The measured initial and final entropy values were both 8.0301, resulting in an observed entropy reduction of 0.00%.

This result is reported as an empirical property of the implemented simulation. It is not interpreted as a universal mathematical guarantee that entropy decreases after every realized observation.

A.7. Navigation Error

Angular navigation error was calculated as the great-circle separation between the true spacecraft direction and the estimated direction.

For true spherical coordinates (θ_t, ϕ_t) and estimated coordinates (θ_e, ϕ_e) , the angular separation can be calculated from

$$\Delta\theta = \cos^{-1} [\sin \theta_t \sin \theta_e + \cos \theta_t \cos \theta_e \cos(\phi_t - \phi_e)]. \quad (35)$$

The resulting error was expressed in degrees.

The evaluation reported mean, median, standard deviation, and 95th-percentile angular errors.

A.8. Noise Model

Measurement noise was introduced to investigate the robustness of the navigation framework under increasingly uncertain CMB observations. Noise levels were expressed relative to the RMS of the simulated CMB signal.

The evaluated noise conditions ranged from 5% to 50% of the signal RMS. The same general observation-processing pipeline was maintained across noise conditions so that changes in navigation performance could be attributed to the simulated measurement uncertainty.

A.9. Statistical Evaluation

The navigation errors produced by nearest-neighbour and Bayesian inference were compared using the Wilcoxon signed-rank test [7]. This non-parametric paired test was selected because the comparison is based on corresponding navigation cases rather than independent samples.

The significance threshold was set to

$$\alpha = 0.05. \quad (36)$$

The standardized magnitude of the difference was additionally quantified using Cohen’s d [8].

For the evaluated experiment, the Wilcoxon test produced

$$p = 0.4516, \quad (37)$$

while the calculated effect size was

$$d = -0.0087. \quad (38)$$

Therefore, the null hypothesis was not rejected and no statistically significant performance difference was established between the two methods.

A.10. Experimental Software Environment

The computational framework was implemented in Python. HEALPix operations and spherical map processing were performed using `healpy` [4]. Numerical calculations and machine-learning-based feature analysis were performed using standard scientific Python libraries.

The computational workflow consisted of the following stages:

1. Generate the full-sky CMB temperature field.
2. Extract local CMB patches.
3. Compute the twenty-dimensional fingerprints.
4. Construct the reference fingerprint database.
5. Build the nearest-neighbour index.
6. Generate noisy spacecraft observations.

7. Evaluate nearest-neighbour localization.
8. Perform sequential Bayesian updates.
9. Calculate posterior entropy and localization error.
10. Evaluate noise sensitivity.
11. Evaluate observation-count dependence.
12. Tune the likelihood parameter σ .
13. Analyze fingerprint discriminability.
14. Calculate feature importance.
15. Perform statistical comparison.

A.11. Reproducibility Considerations

The experiments were designed as a simulation-based proof-of-concept. Reproducibility depends on the CMB power-spectrum parameters, map resolution, random-number generation, reference-position sampling, feature-extraction procedure, noise realization, likelihood parameter, and test-position selection.

For strict reproducibility, future releases of the CryoNavigation implementation should record the random seeds, complete cosmological parameter set, software versions, hardware configuration, and exact generated reference database. The complete experimental pipeline should also be evaluated using multiple independently generated CMB realizations and larger test sets.

A.12. Supplementary Result Summary

For convenience, the principal numerical findings are stated below without reproducing the tables and figures already presented in the main paper.

- The simulated CMB map contained 49,152 pixels at $\text{NSIDE} = 64$.
- The reference database contained 3,072 candidate positions.
- Each candidate position was represented by a twenty-dimensional fingerprint.
- The nearest-neighbour mean angular error was 89.8200° .
- The Bayesian mean angular error was 90.1565° .

- The change in mean error was approximately -0.37% .
- The Bayesian standard deviation was 34.6021° , compared with 42.2774° for nearest-neighbour matching.
- The Bayesian 95th-percentile error was 137.7878° , compared with 156.2222° for nearest-neighbour matching.
- Eight sequential observations produced an entropy reduction of 0.00% .
- The Wilcoxon signed-rank test produced $p = 0.4516$.
- Cohen's d was -0.0087 .
- Bayesian inference improved the error in 53.3% of evaluated cases.
- The Pearson correlation between angular separation and fingerprint distance was 0.0477 .
- The estimated angular resolution limit was 34.67° .

These supplementary details support reproducibility while keeping the main manuscript focused on the research motivation, methodology, experimental findings, and scientific interpretation.